

PROJECT DEVELOPMENT REPORT

Vibe Coding & AWS Elastic Beanstalk Deployment | IBM & Bharat Cares

AWS Elastic Beanstalk: <http://studymate-ai-prod-env.eba-6yuyfhhu.ap-southeast-2.elasticbeanstalk.com>

Live HTTPS Public URL: <https://studymate-ai-5z50.onrender.com>

1. Application Overview & Tech Stack

StudyMate.AI is a full-stack, AI-powered study platform built using modern Vibe Coding techniques, containerized with Docker, and deployed on AWS Elastic Beanstalk & Render Cloud Platform.

Tech Stack Breakdown:

- Frontend: React 19, Vite, Tailwind CSS v4, Motion animations, Lucide React icons.
- Backend: Node.js 20, Express.js framework, Esbuild bundler, Mammoth document parser.
- AI Model: Google Gemini 3.6 Flash (@google/genai SDK).
- Containerization: Docker (multi-stage Dockerfile based on Node.js 20-slim, Dockerrun.aws.json v1).
- Cloud Infrastructure: AWS Elastic Beanstalk & Render Cloud Platform.

2. Prompting Strategy & Frameworks Used

Persona & System Prompting

The AI is assigned a pedagogical persona ('Expert Academic Tutor') instructed to maintain clarity, encouraging tone, and structured formatting.

Structured Output Schema (JSON)

For Quiz Generation, prompts enforce a rigid JSON schema enforcing question text, options array, correct option index, and detailed explanation to ensure flawless UI parsing.

3. Phase-by-Phase Development Summary

Phase 1: Design & UI System

Implemented frosted-glass aesthetic with dark mode and mobile responsiveness.

Phase 2: Backend & LLM Integration

Built Express API endpoints (/api/simplify, /api/tutor, /api/quiz) ensuring Gemini API keys remain strictly secure on the server side.

Phase 3: Docker Containerization

Created multi-stage Dockerfile packaging the full-stack app into a lightweight, isolated Docker container.

Phase 4: AWS Elastic Beanstalk & HTTPS Deployment

Deployed AWS Elastic Beanstalk environment on Docker Platform and established live HTTPS public access.

4. Application Architecture

Client Browser % % (HTTPS/Port 443)% % %° Reverse Proxy / Cloud Infrastructure % % %° Docker Container (Node.js App) % % (API Key / SDK)% % %° Google Gemini 3.6 API

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5. Challenges Encountered & Resolutions

Challenge 1: Multi-Stage Docker Build for Cloud Engines

Configured multi-stage Dockerfile compiling Vite & Esbuild inside container during build phase.

Challenge 2: Public HTTPS Encryption Compliance

Configured Render Cloud HTTPS deployment to complement AWS Elastic Beanstalk, guaranteeing 100% SSL encryption compliance.

Challenge 3: Environment Variable Security

Ensured API keys are never bundled in client code. Configured GEMINI_API_KEY as server-side Environment Variable.

6. Key Learnings & Reflection

This project demonstrated end-to-end Vibe Coding methodology—architecting, containerizing with Docker, and deploying a functional AI web application on AWS Elastic Beanstalk & HTTPS Cloud infrastructure with zero cost and maximum security.